E – LEARNING PLATFORM FOR HEARING IMPAIRED STUDENTS

Group: 25-26J-103

INTRODUCTION: E-LEARNING PLATFORM FOR HEARING IMPAIRED STUDENTS

PROBLEM STATEMENT & STAKEHOLDERS

Stakeholders :

- Primary: Hearing impaired students
- Secondary: Teachers, sign language instructors
- Decision makers: Deaf School administrators

Problem:

Hearing-impaired students lack access to culturally relevant, multisensory, real-time digital learning tools that support their unique communication and emotional needs.

Why it matters:

This gap limits educational engagement, emotional development through music, sign language mastery, and communication skills . Directly affecting academic outcomes and quality of life.

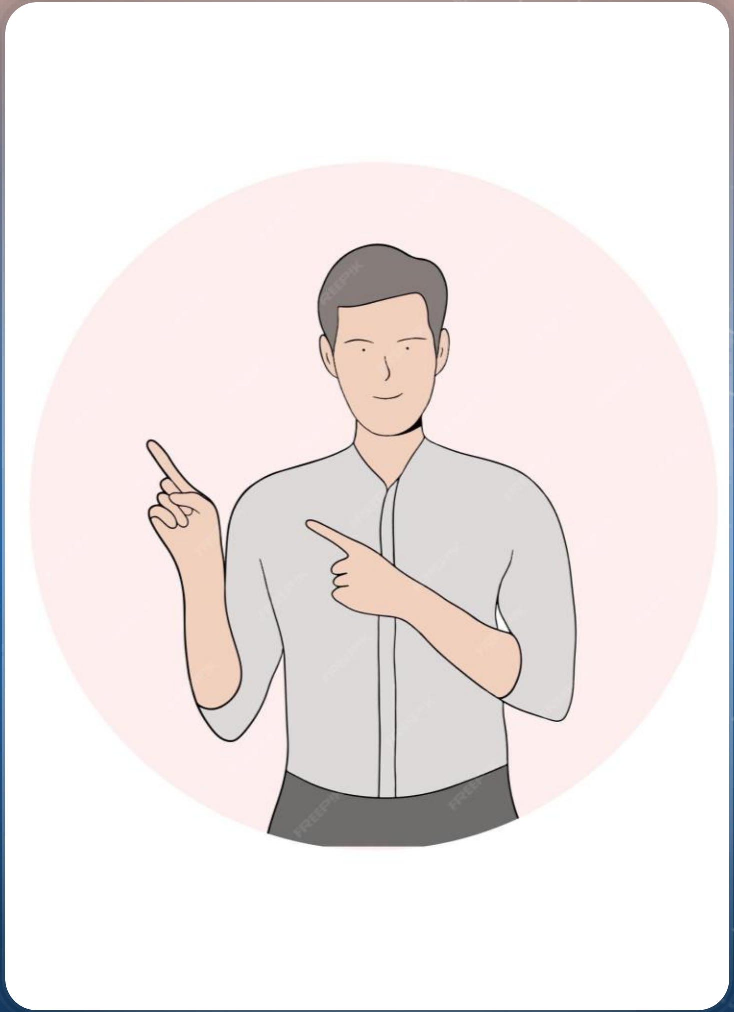
RESEARCH GAP & OBJECTIVES

Gap:

- No synchronized multisensory music experience for Sinhala/English music
- No real-time SLSL gesture correction platform
- No AI-powered Sinhala lip-reading assistant with 3D avatar feedback
- No emotion-aware, visual-only learning monitoring for hearing-impaired students

Main Objective:

To develop an integrated AI-powered e-learning platform that enables hearing-impaired students to learn through multisensory feedback, sign language interaction, lip-reading, and emotion-adaptive content with measurable improvements in engagement and communication skills.



SOLUTION OVERVIEW

Solution Summary:

A four-component AI-driven e-learning platform designed specifically for hearing-impaired students, combining haptic feedback, sign language learning, lip-reading, and emotion monitoring.

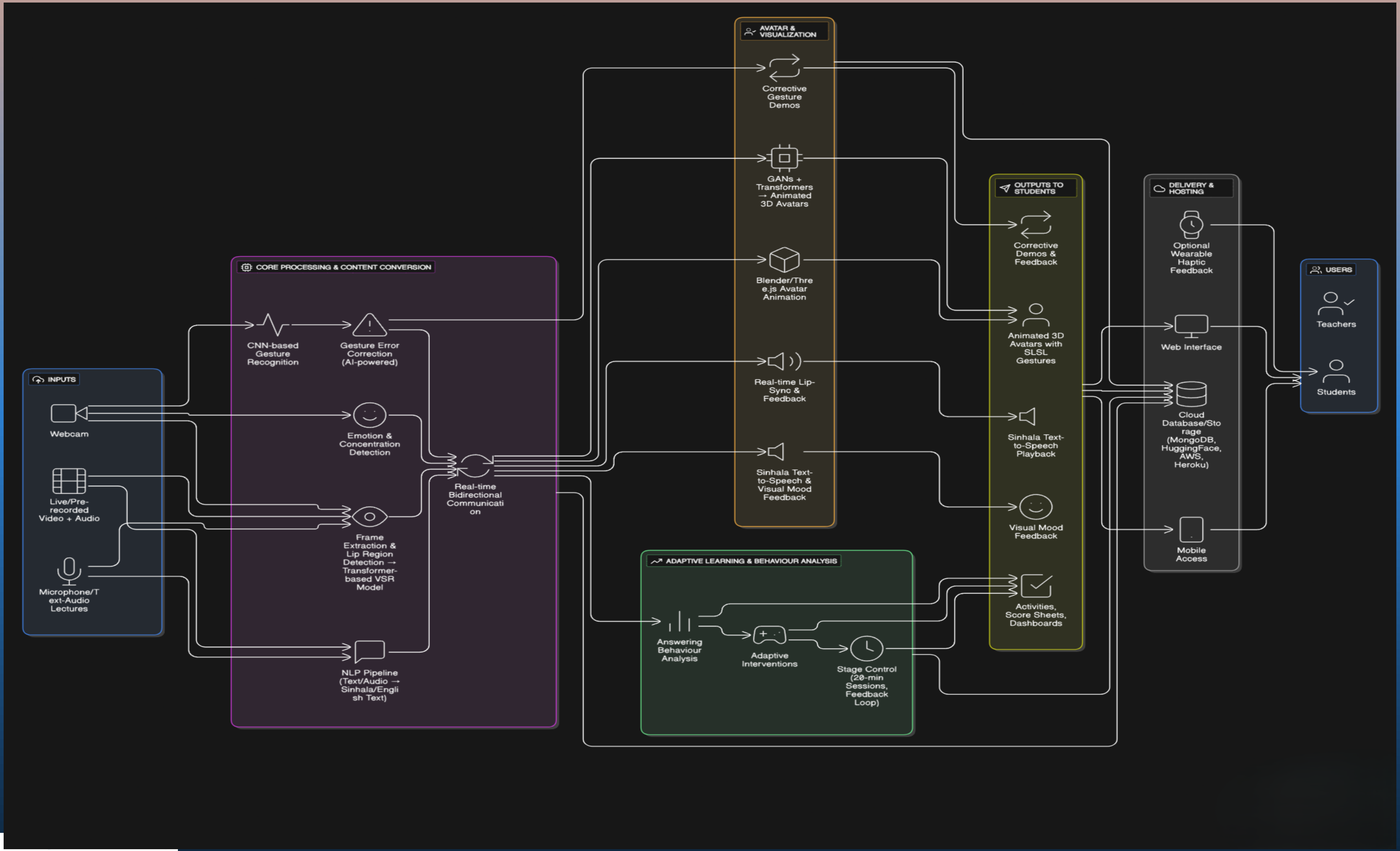
Component	Owner	Feature
Emotion-Aware Music using multisensory	Viduranga W R A V (IT22255488)	Emotion Recognition System+Music Transcription System +Haptic Feedback Hardware Module + AI Chatbot & Summarization System+Visualization System
SLSL Gesture Correction	Jayasinha J.A.N.Y. (IT22546852)	Webcam-based sign recognition + real-time corrective feedback
Sinhala Lip-Reading + 3D Avatar	Samaraweera T.N (IT22143754)	Neural network lip-reading + Blender 3D avatar animation
Emotion Monitoring	Hettiarachchi D.S.W (IT22561466)	Facial emotion detection + stage-based learning optimization

NON-FUNCTIONAL FOCUS

- Real-time performance
- Accessibility
- Scalability

➢ **Current Completion Status: ~92%**

SYSTEM ARCHITECTURE DIAGRAM



COMMERCIALIZATION AND OVERALL PROJECT

Potential Market:

- Educational institutions (special schools, universities).
- Healthcare & rehabilitation centers for hearing-impaired users.
- Public service kiosks & customer service platforms.

Revenue Model:

- Freemium web platform with premium enterprise features.
- Licensing to schools, hospitals, and NGOs.
- Paid API access for third-party integrations.

Scalability:

- Can expand to other languages & regions.
- Integration with AR/VR devices for enhanced interaction.

Long-term Vision:

- Commercial product with subscription-based model.
- Continuous AI model improvements with user feedback.

Category	Estimated Cost (LKR)
Dataset Preparation & Annotation	25,000
Internet cost	10,000
Cloud Computing & Hosting	20,000
Development Tools	15,000
Database & Storage	15,000
Testing & Evaluation Tools	10,000
Miscellaneous & Contingency	15,000